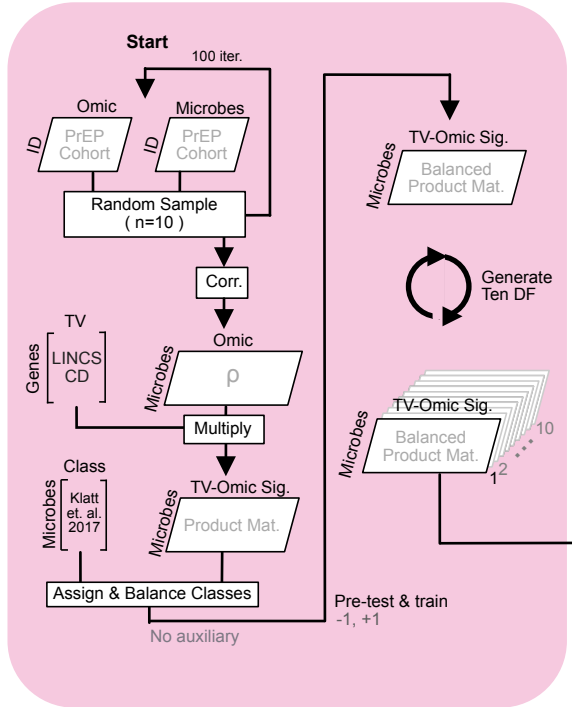


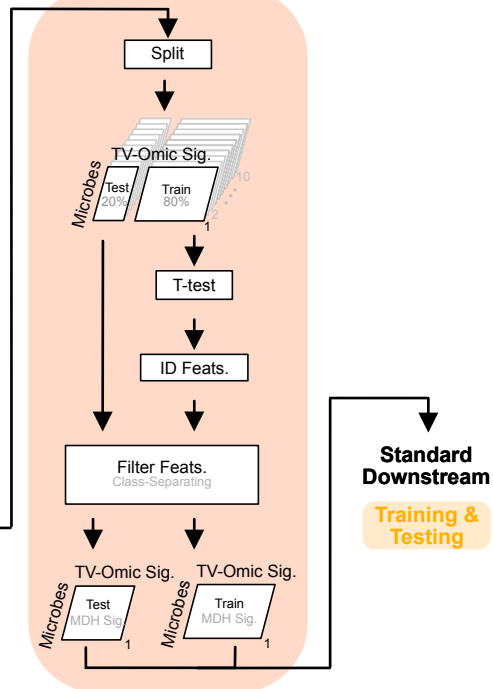
A

TV-VagGene & TV-VagProt

Data Bootstrapping & Pre-processing



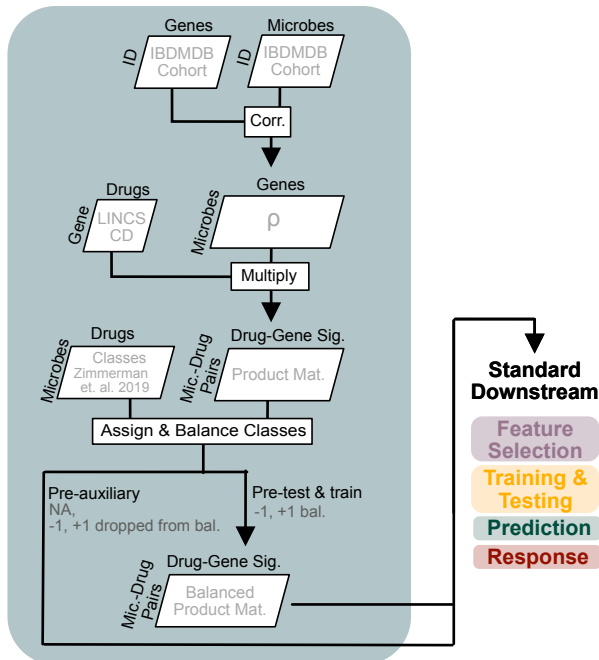
Feature Selection



B

Multi-GutV1 & Multi-GutV2

Data Pre-processing



Supplementary Figure 2: Model specific pre-processing steps. A) TV-VagGene and TV-VagProt pre-processing involves bootstrapping omic data and microbiome composition ($n = 10$, 100 iterations) to generate ρ which are multiplied by the TV-treated LINC CD. This is repeated 10 times to generate 10 bootstrapped balanced product matrices. Feature selection identifies 500 class-separating features with the lowest adjusted p-values per data frame, features shared across the 10 data frames will define the MDH signatures. No auxiliary data is generated since novel predictions are completed on direct patient correlations. B) Multi-GutV1 and Multi-GutV2 ρ values are multiplied by each drug-treated LINC CD, generating a microbe-drug pair product matrix. Classes are assigned from *in vitro* data, which are defined by the *percent consumed threshold* and *microbe proportion threshold*.